Project Proposal: Real-Time Urban Air Quality Monitoring & Analytics Dashboard

1 Project Title

EcoStream: Real-Time Urban Air Quality Monitoring & Analytics Dashboard

2 Problem Statement

Rapid urban growth has significantly increased air pollution levels in cities around the world. Although environmental agencies publish air quality data, it is often presented in static reports or daily summaries that are not easy for the general public to interpret. As a result, people lack access to simple, real-time information about the air they breathe.

This project aims to solve that issue by developing an interactive dashboard that collects and visualizes live environmental data. By presenting pollutant levels in an understandable and dynamic format, the system will help citizens, researchers, and urban planners monitor pollution patterns, identify high-risk areas, and make informed decisions about outdoor activities and environmental planning.

3 Assigned Category

Real-Time Data Analytics Dashboard

4 Key Features

1. **Live Multi-Sensor Data Integration**
   The system will continuously gather air quality readings such as PM2.5, PM10, CO2, temperature, and humidity from either real IoT devices or a simulated data stream. Data will refresh approximately every 30–60 seconds to ensure near real-time monitoring.
2. **Interactive Geospatial Heatmap**
   An interactive map will display sensor locations using color-coded markers that reflect the Air Quality Index (AQI). Users can quickly identify pollution hotspots across different areas of the city.
3. **Real-Time Pollutant Trend Graphs**
   Dynamic time-series charts will show pollutant concentration trends over selectable time intervals such as the last hour, six hours, or 24 hours. These graphs will update automatically as new data arrives.
4. **Health Advisory KPI Cards**
   Summary indicators will convert raw pollutant readings into clear health guidance

messages. For example, users may see alerts like “Air Quality is Good” or “Sensitive Groups Should Limit Outdoor Activity.”

5. **Smart Threshold Alerts**
   The dashboard will include a customizable alert system that highlights warning messages when pollutant levels exceed recommended safety standards set by recognized health organizations.
6. **Comparative Location Analysis**
   Users will be able to compare data from two different sensor locations side-by-side, helping them determine which area currently has better air quality.
7. **Automated AQI Calculation**
   The system will automatically compute the Air Quality Index using official environmental guidelines. This ensures accurate and standardized interpretation of pollutant data.
8. **.Historical Trend Comparison**
   Current readings can be displayed alongside previous week averages to help users detect unusual spikes or improvements in air quality.

5 Libraries You Plan to Use

- Requests / HTTPX – For retrieving environmental data from APIs or simulated sensor streams.
- Pandas – For organizing, cleaning, and analyzing time-series data.
- NumPy – For performing mathematical computations required in AQI calculations.
- Plotly and Dash – For building the interactive dashboard and visual components, including charts and maps.
- SQLite with SQLAlchemy – For storing recent historical data to support comparison and trend analysis features.
- Asyncio – For handling asynchronous data collection without interrupting dashboard performance.

6 System Scope

What WILL be included:

- A backend data pipeline that retrieves environmental readings every 30–60 seconds.
- An interactive web-based dashboard featuring maps, graphs, and health indicators.
- An alerting mechanism that monitors pollutant thresholds in real time.
- A realistic data simulation system if live sensor hardware is not available.

What will NOT be included:

- Physical IoT sensor hardware installation.
- Predictive machine learning models for future pollution forecasting.

- User account creation or login functionality.
- Development of a standalone mobile application.

7 Clear Success Criteria

- The project will be considered successful if:

- The dashboard updates within 10 seconds of receiving new data.
- At least 10–15 sensor locations are accurately displayed on the interactive map.
- AQI calculations align with official environmental standards within a 1% margin of error.
- Users can switch between multiple pollutants (e.g., PM2.5, CO2, temperature) without performance issues.
- The system runs continuously for at least four hours without crashes or interruptions during demonstration.